The minimax rate for Darcy's inverse problem near a constant conductivity

Abstract

For a sufficiently small, fixed Hölder–Zygmund ball of conductivities around the constant function 1, the minimax squared L^2-risk is, up to logarithmic factors,

$$n^{-2\alpha/(2\alpha+d+3)}.$$

We prove matching polynomial upper and lower bounds for every $d \geq 2$ and $\alpha > 1$. The upper bound uses a regularized reconstruction along gradient trajectories, together with a variance estimate that exploits integration along these trajectories. The lower bound uses perturbations that oscillate in the $d-1$ angular directions. No boundary condition on the conductivity is imposed.

1 Statement and conventions

Let $D = B(0,1) \subset \mathbb{R}^d$. We interpret $C^s(D)$ as the Hölder–Zygmund restriction space $B^s_{\infty,\infty}(D)$. Any fixed usual equivalent norm may be used; the constants below, including the admissible radius, may depend on this choice.

White noise means the isonormal experiment

$$Y_n(\varphi) = \langle u_f, \varphi \rangle_{L^2(D)} + n^{-1/2}\mathbb{W}(\varphi), \qquad \varphi \in L^2(D),$$

where $\mathbb{E}[\mathbb{W}(\varphi)\mathbb{W}(\psi)] = \langle \varphi, \psi \rangle$. Write

$$\mathcal{F}(\alpha, r) = \{f : \|f - \mathbf{1}\|_{C^\alpha(D)} \leq r\}, \qquad R_n(\alpha, r) = \inf_{\widehat{f}} \sup_{f \in \mathcal{F}(\alpha,r)} \mathbb{E}_f \|\widehat{f} - f\|^2_{L^2(D)}.$$

Theorem 1.1. *Fix $d \geq 2$ and $\alpha > 1$. There exists $r_* > 0$ such that, for each fixed $r \in (0, r_*]$, there are constants $c, C > 0$ and n_0 for which*

$$c\, n^{-2\alpha/(2\alpha+d+3)} \leq R_n(\alpha, r) \leq C(\log n)^4 n^{-2\alpha/(2\alpha+d+3)}, \qquad n \geq n_0.$$

If $\alpha \notin \mathbb{N}$, the logarithmic factor in this upper bound can be omitted. In particular, the answer up to logarithmic factors is

$$\boxed{R_n(\alpha, r) = \eqsim n^{-2\alpha/(2\alpha+d+3)}.}$$

For a smooth function a, the notation

$$\|a\|_{k,\infty;\Omega} = \max_{|\beta| \leq k} \|\partial^\beta a\|_{L^\infty(\Omega)}$$

denotes the classical derivative norm, not an integer-index Zygmund norm. Domain subscripts will be suppressed when unambiguous. Constants denoted by C may change from line to line, but are independent of n, the bandwidth, and the conductivity in the specified parameter class.

2 Analytic preliminaries

2.1 Function spaces and the forward equation

We use the Littlewood–Paley characterization

$$\|a\|_{C^s(\mathbb{R}^d)} \asymp \sup_{j\ge 0} 2^{js}\|\Delta_j a\|_\infty \tag{1}$$

for an inhomogeneous smooth dyadic decomposition. The equivalence with classical Hölder spaces at noninteger positive indices, and the differentiation properties of these spaces, are given in [3, Chapter 13, Section 8].

We also use a common bounded extension operator from a smooth bounded domain to $\mathbb{R}^d$, bounded on all the finitely many Besov spaces under consideration. Such an operator exists even for Lipschitz domains; see [2].

Here are some consequences, with the relevant arguments recalled. Differentiation maps C^s boundedly to C^{s-1}. For $s > 0$, C^s is a multiplication algebra:

$$\|ab\|_{C^s} \le C_s\|a\|_{C^s}\|b\|_{C^s}. \tag{2}$$

Indeed, decompose the product into low–high, high–low, and comparable-frequency products. The low–high contribution to its j-th dyadic block is bounded by $C\|a\|_\infty 2^{-js}\|b\|_{C^s}$; interchange a, b for the high–low contribution. For comparable frequencies $k \ge j - C$, the bound is

$$C\|a\|_{C^s}\|b\|_{C^s}\sum_{k\ge j-C} 2^{-2ks} \le C_s 2^{-js}\|a\|_{C^s}\|b\|_{C^s}.$$

This proves (2). Extension and restriction give the same facts on D.

We shall also use the real interpolation identity

$$(C^{s_0}, C^{s_1})_{\theta,\infty} = C^{(1-\theta)s_0+\theta s_1}, \qquad s_0 < s_1, \quad 0 < \theta < 1. \tag{3}$$

For completeness, on $\mathbb{R}^d$ this follows by splitting the dyadic series at a level J. If $s = (1-\theta)s_0 + \theta s_1$, the low and high parts satisfy

$$\|a_{\le J}\|_{C^{s_1}} \le C2^{J(s_1-s)}\|a\|_{C^s}, \qquad \|a_{>J}\|_{C^{s_0}} \le C2^{-J(s-s_0)}\|a\|_{C^s}.$$

Taking $2^{J(s_1-s_0)} \asymp t^{-1}$ bounds the K-functional by $Ct^\theta\|a\|_{C^s}$. Conversely, applying Δ_j to any decomposition $a = a_0 + a_1$ gives

$$\|\Delta_j a\|_\infty \le C2^{-js_0}K(2^{-j(s_1-s_0)}, a; C^{s_0}, C^{s_1}),$$

which proves the reverse bound. A common extension operator transfers the identity to D.

Lemma 2.1 (Uniform perturbative regularity)**.** *Put*

$$u_\circ(x) = \frac{|x|^2 - 1}{2d}.$$

For sufficiently small $r_* > 0$, every $f \in \mathcal{F}(\alpha, r_*)$ satisfies $1/2 \le f \le 3/2$, and the Dirichlet problem

$$\operatorname{div}(f\nabla u) = 1, \qquad u|_{\partial D} = 0$$

has a unique weak solution. Moreover,

$$\|u_f - u_\circ\|_{C^{\alpha+1}(D)} \le C\|f - \mathbf{1}\|_{C^\alpha(D)}. \tag{4}$$

Proof. Let $\mathcal{P}F$ be the zero-Dirichlet solution of $\Delta w = \operatorname{div} F$. For every $s > 1$,

$$\|\mathcal{P}F\|_{C^{s+1}(D)} \le C_s \|F\|_{C^s(D)}. \tag{5}$$

At noninteger s, this is the classical global Schauder estimate and solvability theorem, including higher-order regularity; see [1, Theorems 6.6, 6.14 and 6.19]. At integer s, apply these bounds at two surrounding noninteger indices and use (3). Thus (5) holds also at the Zygmund indices.

Writing $u = u_\circ + w$, the desired equation becomes

$$w = -\mathcal{P}\big((f-1)\nabla u_\circ\big) - \mathcal{P}\big((f-1)\nabla w\big).$$

By (2) and (5), the second term is a linear operator on the zero-trace subspace of $C^{\alpha+1}(D)$ with norm at most $C\|f-1\|_{C^\alpha}$. Choosing r_* so that this norm is at most $1/2$, the Neumann series gives a solution and (4).

The embedding $C^\alpha(D) \hookrightarrow C^0(\overline{D})$ allows us to decrease r_* so that $1/2 \le f \le 3/2$. Uniqueness of a weak solution follows by testing the equation for the difference of two solutions against that difference: $\int_D f|\nabla w|^2 = 0$. Finally, the constructed solution has zero trace and belongs to $H^1_0(D)$, so it is that unique weak solution. □

2.2 Smoothing and a product cancellation

Set

$$m_0 = \lceil \alpha \rceil - 1.$$

Fix a real $K \in C^\infty_c(B(0,1))$ such that

$$\int K = 1, \qquad \int x^\beta K(x)\,dx = 0 \quad (1 \le |\beta| \le 2m_0 + 2). \tag{6}$$

Such a kernel can be obtained from a finite linear combination of distinct dilations of a smooth unit-integral bump: the required coefficients solve a Vandermonde system in the dilation parameters. Positivity of K is neither required nor asserted.

For $0 < h < 1/8$, put

$$K_h(x) = h^{-d}K(x/h), \qquad \Omega_h = B(0, 1-2h), \qquad S_h a(x) = \int_D K_h(x-y)a(y)\,dy, \quad x \in \Omega_h,$$

and let $L_h = \log(e/h)$.

Lemma 2.2 (Smoothing and commutator bounds)**.** *Suppose $a, b \in C^\alpha(D)$ have bounded norms. Uniformly on Ω_h,*

$$\|S_h a - a\|_\infty \le C h^\alpha L_h, \tag{7}$$

$$\big\|\partial^\gamma\big((S_h a)(S_h b) - S_h(ab)\big)\big\|_\infty \le C h^{\alpha+1-|\gamma|} L_h^2, \qquad |\gamma| \le 2. \tag{8}$$

If $\alpha \notin \mathbb{N}$, all factors L_h in these two bounds may be replaced by 1. The statements hold componentwise for vector-valued b.

Proof. We first record the local Taylor remainder

$$\sup_{|y-x|\le h} |a(y) - P_{a,x}(y)| \le C h^\alpha L_h, \tag{9}$$

where $P_{a,x}$ is the Taylor polynomial of degree m_0. Its coefficients are uniformly bounded. At noninteger α, this is the ordinary Hölder Taylor estimate, without L_h. At integer α, derivatives of order $\alpha-1$ have modulus of continuity bounded by $Ct\log(e/t)$. To see the latter directly from (1), split their dyadic series at $2^J \asymp t^{-1}$: the low-frequency increments sum to $C\sum_{j\le J} t$, and the high-frequency tail is at most $C\sum_{j>J} 2^{-j}$. Taylor's integral remainder then gives (9). Extension to $\mathbb{R}^d$ justifies this argument uniformly near the boundary.

Polynomial reproduction from (6) immediately proves (7). To prove (8), write

$$Q_h(a,b) = (S_h a)(S_h b) - S_h(ab).$$

Fix x and write $a = P_a + R_a$, $b = P_b + R_b$, using the Taylor polynomials at x. Keep these polynomials fixed when differentiating at x. Polynomial reproduction gives $Q_h(P_a, P_b) = 0$. Also $Q_h(a,c) = 0$ for every constant c.

On $B(x,h)$, the remainders satisfy $|R_a|+|R_b| \le Ch^\alpha L_h$, whereas $|P_a - P_a(x)|+|P_b - P_b(x)| \le Ch$. Differentiating the convolution kernels gives, for $|\gamma| \le 2$,

$$|\partial^\gamma Q_h(R_a, P_b)(x)| + |\partial^\gamma Q_h(P_a, R_b)(x)| \le Ch^{\alpha+1-|\gamma|}L_h.$$

For example, in the first expression subtract the constant $P_b(x)$; the differentiated convolution of $R_a(P_b - P_b(x))$ has precisely this bound. In the product term, a derivative falling on the polynomial is bounded by Ch^{1-k} at derivative order $k \ge 1$; the undifferentiated polynomial vanishes at x. Finally,

$$|\partial^\gamma Q_h(R_a, R_b)(x)| \le Ch^{2\alpha-|\gamma|}L_h^2.$$

Since $\alpha > 1$, these inequalities prove (8). The same proof without logarithms applies at noninteger α. □

Lemma 2.3 (Completion of a thin boundary strip)**.** *For all sufficiently small h, there is a linear map $E_h : L^2(\Omega_h) \to L^2(D)$ such that*

$$\|E_h a\|_{L^2(D)} \le C\|a\|_{L^2(\Omega_h)}, \tag{10}$$

$$\|E_h(f|_{\Omega_h}) - f\|_{L^2(D)} \le C\|f\|_{C^\alpha(D)}h^\alpha L_h. \tag{11}$$

At noninteger α, L_h can be omitted.

Proof. Let $q = m_0 + 1$ and $\lambda_j = (-1)^{j+1}\binom{q}{j}$, $1 \le j \le q$. Then

$$\sum_{j=1}^{q} \lambda_j = 1, \qquad \sum_{j=1}^{q} \lambda_j j^k = 0, \quad 1 \le k \le m_0.$$

Define $E_h a = a$ on Ω_h, and, for $x \in D \setminus \Omega_h$, define

$$E_h a(x) = \sum_{j=1}^{q} \lambda_j a((1-3jh)x).$$

For $h < 1/(6q)$, all arguments on the right belong to Ω_h and all dilation factors are at least $1/2$. A change of variables proves (10). Taylor expansion of f at x, together with the displayed moment identities and (9), proves (11). □

3 Inversion along gradient trajectories

For a smooth function v, define

$$T_v a = \operatorname{div}(a\nabla v) = \nabla v \cdot \nabla a + (\Delta v)a.$$

Lemma 3.1 (Transport inverse)**.** *Let $\Omega = B(0,R)$, $3/4 \le R \le 1$, and suppose v is smooth on a neighborhood of $\overline{\Omega}$, with*

$$\lambda I \le D^2 v \le \Lambda I, \qquad \nabla v \cdot \nu > 0 \text{ on } \partial\Omega, \tag{12}$$

where $0 < \lambda \le \Lambda$. There is a unique critical point $p \in \Omega$. Let $z_t(x)$ solve

$$\frac{d}{dt} z_t(x) = -\nabla v(z_t(x)), \qquad z_0(x) = x.$$

This flow exists in $\overline{\Omega}$ for all $t \ge 0$, and

$$e^{-\Lambda t}|x-p| \le |z_t(x) - p| \le e^{-\lambda t}|x-p|. \tag{13}$$

For smooth g, the unique bounded smooth solution of $T_v a = g$ is

$$P_v g(x) = \int_0^\infty \exp\left\{ -\int_0^t \Delta v(z_s(x))\, ds \right\} g(z_t(x))\, dt. \tag{14}$$

With $c_0 = d\lambda$, it satisfies

$$\|P_v g\|_\infty \le c_0^{-1} \|g\|_\infty, \tag{15}$$

$$\|P_v g\|_2 \le 2c_0^{-1} \|g\|_2, \tag{16}$$

$$\|P_v g\|_{H^1} \le C\big\{ \|g\|_{H^1} + (1 + \|D^3 v\|_\infty)\|g\|_2 \big\}. \tag{17}$$

The constants in the last estimate depend only on d, λ, Λ.

Proof. Strict convexity gives at most one critical point. A minimum on $\overline{\Omega}$ cannot occur at the boundary, because the inward direction decreases v, so this critical point exists in Ω. The vector field $-\nabla v$ points inward at the boundary; the usual first-exit argument and local existence for a smooth vector field therefore give a global backward-gradient flow in $\overline{\Omega}$.

For $z \in \Omega$,

$$\lambda |z-p|^2 \le (z-p) \cdot \nabla v(z) \le \Lambda |z-p|^2.$$

Applying this to the derivative of $|z_t - p|$ proves (13).

Since $\Delta v \ge c_0$, the integral in (14) converges absolutely and gives (15). It is smooth up to the boundary. Here is a justification of differentiating the infinite-time integral. The first variational equation gives $\|D_x z_t\| \le e^{-\lambda t}$. Inductively, the equation for $D_x^k z_t$ has homogeneous part $-D^2 v(z_t) D_x^k z_t$, while each forcing term is a product of at least two lower-order flow derivatives. Variation of constants therefore gives $\|D_x^k z_t\| \le C_k e^{-\lambda t}$. Derivatives of $\int_0^t \Delta v(z_s)\, ds$ of every positive order are consequently bounded uniformly in t. Every differentiated integrand is thus bounded by $C_k e^{-c_0 t}$.

The flow semigroup property, or differentiation after splitting the integral at a finite time, gives $T_v P_v g = g$. If $T_v a = 0$, differentiation along the flow shows

$$a(x) = \exp\left\{ -\int_0^t \Delta v(z_s(x))\, ds \right\} a(z_t(x)).$$

Letting $t \to \infty$ proves uniqueness among bounded solutions.

For $a = P_v g$, integration by parts gives

$$\int_\Omega a T_v a = \frac{1}{2}\int_{\partial\Omega} (\nabla v \cdot \nu) a^2 + \frac{1}{2}\int_\Omega (\Delta v) a^2 \geq \frac{c_0}{2}\|a\|_2^2.$$

This proves (16). Differentiating the equation gives

$$T_v(\nabla a) + D^2 v\, \nabla a = \nabla g - a \nabla \Delta v.$$

The same integration by parts, now for a vector field, yields

$$(c_0/2 + \lambda)\|\nabla a\|_2 \leq \|\nabla g\|_2 + \|\nabla \Delta v\|_\infty \|a\|_2.$$

Together with (16), this proves (17). □

4 Gaussian smoothing and the decisive variance estimate

Define

$$Z_h(x) = n^{-1/2}\mathbb{W}(K_h(x - \cdot)), \qquad x \in \mathbb{R}^d,$$

where the white-noise argument is restricted to D.

Lemma 4.1 (Gaussian smoothing bounds)**.** *For every fixed nonnegative integer k and $p \geq 1$,*

$$\mathbb{E}\|Z_h\|_{k,\infty;\Omega_h}^p \leq C_p n^{-p/2} h^{-p(d/2+k)} L_h^{p/2}, \tag{18}$$

$$\mathbb{E}\|Z_h\|_{H^k(\Omega_h)}^2 \leq C n^{-1} h^{-d-2k}, \tag{19}$$

$$\mathbb{E}\|Z_h\|_{H^k(\Omega_h)}^4 \leq C n^{-2} h^{-2d-4k}. \tag{20}$$

Moreover, for $t \geq 0$,

$$\mathbb{P}\left(\|Z_h\|_{k,\infty;\Omega_h} > C n^{-1/2} h^{-d/2-k}\sqrt{L_h + t}\right) \leq C e^{-t}. \tag{21}$$

Proof. The white-noise isometry gives

$$\mathbb{E}|\partial^\beta Z_h(x)|^2 \leq C n^{-1} h^{-d-2|\beta|}.$$

Integrating proves (19). For centered jointly Gaussian variables X, Y, $\mathbb{E}(X^2Y^2) \leq 3(\mathbb{E}X^2)(\mathbb{E}Y^2)$; double integration proves (20).

The field has a smooth version. Indeed, after multiplication by a fixed smooth cutoff, its weak derivatives of every order have finite expected squared L^2-norm. Almost surely it therefore belongs to every integer Sobolev space locally. Fourier inversion and Cauchy–Schwarz imply $H^s(\mathbb{R}^d) \hookrightarrow C^k$ for $s > k + d/2$, which gives the assertion.

For a derivative of order k, let $A = n^{-1/2}h^{-d/2-k}$. Its Gaussian increment metric is bounded by

$$CA \min\{1, |x - y|/h\}.$$

On a fixed cube containing Ω_h, use successive grids of mesh $h2^{-j}$. The initial grid has at most Ch^{-d} points, and the j-th grid has at most $Ch^{-d}2^{jd}$ points. Join each point to a nearest point of the preceding grid. Gaussian tail bounds and a union bound show that, outside an event of probability

Ce^{-t}, the initial values are bounded by $CA\sqrt{L_h + t}$, and every increment at level $j \geq 1$ is bounded by

$$CA2^{-j}\sqrt{L_h + t + j}.$$

The constants can be chosen so that the failure probabilities at level j sum to Ce^{-t}. Summing the increments and using continuity proves (21). A finite union handles all derivatives of order at most k. Integrating the tail proves (18). □

The next lemma is responsible for the denominator $2\alpha + d + 3$.

Lemma 4.2 (Variance after transport inversion)**.** *Suppose v satisfies (12) on Ω_h, and*

$$2d\lambda > \Lambda.$$

If $\|a\|_{1,\infty;\Omega_h} \leq A_0$, then

$$\mathbb{E}\,\|P_v \operatorname{div}(a\nabla Z_h)\|^2_{L^2(\Omega_h)} \leq Cn^{-1}h^{-d-3}. \tag{22}$$

The constant depends only on $d, \lambda, \Lambda, A_0, K$, and not on higher derivatives of v or a.

Proof. For $z \in \Omega_h$, set

$$H_h(z, y) = a(z)\Delta K_h(z - y) + \nabla a(z) \cdot \nabla K_h(z - y).$$

Its $L^2(D)$-norm is at most $Ch^{-d/2-2}$, and its support in y is contained in $B(z, h)$. Consequently,

$$|\langle H_h(z, \cdot), H_h(w, \cdot)\rangle| \leq Ch^{-d-4}\mathbf{1}_{\{|z-w|\leq 2h\}}. \tag{23}$$

Let p be the critical point of v, $c_0 = d\lambda$, and $r = |x - p|$. The representation (14), the white-noise isometry, and (23) give the pointwise variance bound

$$Cn^{-1}h^{-d-4}\int_0^\infty\int_0^\infty e^{-c_0(t+s)}\mathbf{1}_{\{|z_t(x)-z_s(x)|\leq 2h\}}\,ds\,dt.$$

For $s = t + w$, put $r_t = |z_t(x) - p|$. By (13),

$$|z_t - z_{t+w}| \geq r_t - r_{t+w} \geq (1 - e^{-\lambda w})r_t.$$

If $r_t > 4h$, overlap therefore forces $w \leq 4h/(\lambda r_t)$. If $r_t \leq 4h$, the exponential integral in w is bounded by a constant. Thus

$$\int_0^\infty e^{-c_0 w}\mathbf{1}_{\{|z_t-z_{t+w}|\leq 2h\}}\,dw \leq C\min\{1, h/r_t\}.$$

Since $r_t \geq e^{-\Lambda t}r$, the double time integral is at most

$$C\int_0^\infty e^{-2c_0 t}\min\{1, (h/r)e^{\Lambda t}\}\,dt \leq C\min\{1, h/r\}.$$

The last inequality uses $2c_0 > \Lambda$; at $r = 0$ the minimum is interpreted as 1.

Finally, since $d \geq 2$ and $\Omega_h \subset B(p, 2)$,

$$\int_{\Omega_h}\min\{1, h/|x - p|\}\,dx \leq C\left(h^d + h\int_h^2 r^{d-2}\,dr\right) \leq Ch.$$

Integrating the pointwise variance proves (22). All interchanges of integrals are justified by the exponential weights and the uniform L^2-bound on H_h. □

5 The upper bound

Proposition 5.1. *For sufficiently small $r_* > 0$, the upper bound in Theorem 1.1 holds uniformly over $\mathcal{F}(\alpha, r_*)$.*

Proof. Write

$$p_\alpha = 2\alpha + d + 3, \qquad h = n^{-1/p_\alpha}.$$

Only sufficiently large n will be considered.

Construction of the estimator

Form the observable smooth field

$$\widehat{u}_h(x) = Y_n(K_h(x - \cdot)), \qquad x \in \Omega_h.$$

For a fixed true f, use the deterministic notation

$$u = u_f, \qquad v = S_h u, \qquad a = S_h f.$$

Then $\widehat{u}_h = v + Z_h$.

Choose a fixed $\delta > 0$, depending only on d, so small that

$$\|w - u_\circ\|_{2,\infty;\Omega_h} \le \delta$$

implies (12), uniformly for $h < 1/8$, with

$$\lambda = \frac{3}{4d}, \qquad \Lambda = \frac{5}{4d}.$$

Indeed, sufficiently small entrywise perturbations of $D^2 u_\circ = I/d$ give these Hessian bounds, and $\nabla u_\circ \cdot \nu = (1 - 2h)/d$ on $\partial\Omega_h$. These constants also satisfy $2d\lambda > \Lambda$.

Polynomial reproduction and Lemma 2.1 imply

$$\|v - u_\circ\|_{2,\infty} \le C r_*.$$

Also, by moving one derivative onto the kernel,

$$\|a\|_{1,\infty} \le C, \qquad \|a\|_{2,\infty} \le C h^{-1}, \qquad \|v\|_{3,\infty} \le C h^{-1}.$$

Decrease r_* so that the first bound is at most $\delta/2$. Choose a fixed sufficiently large M_0, and define the observable event

$$\mathcal{A}_h = \left\{ \|\widehat{u}_h - u_\circ\|_{2,\infty} \le \delta, \quad h\|\widehat{u}_h\|_{3,\infty} \le M_0 \right\}.$$

Lemma 4.1 implies, uniformly in the parameter class,

$$\mathbb{P}_f(\mathcal{A}_h^c) \le C n^{-3}. \tag{24}$$

To verify this explicitly, take $t = 3 \log n$ in (21). Both the C^2-noise norm and h times the C^3-noise norm are bounded, with this probability, by a constant times

$$n^{-1/2} h^{-d/2-2} \sqrt{\log n} = h^{\alpha - 1/2} \sqrt{\log n} = o(1).$$

On $\mathcal{A}_h$, define

$$\widetilde{f}_h = P_{\widehat{u}_h} \mathbf{1} \quad \text{on } \Omega_h.$$

Let Π denote pointwise projection onto $[1/2, 3/2]$, and set

$$\widehat{f}_n = \begin{cases} \Pi(E_h \widetilde{f}_h), & \mathcal{A}_h, \\ \mathbf{1}, & \mathcal{A}_h^c. \end{cases}$$

This is an estimator based only on the observations. Its measurability follows from the flow formula: finite-time solutions of the smooth ODE depend continuously on the vector field, and the tail of (14) is uniformly exponentially small on $\mathcal{A}_h$. Notice that the auxiliary transport reconstruction does not require a boundary value for $\widetilde{f}_h$.

Deterministic smoothing bias

Set

$$\mathcal{R}_h = 1 - T_v a.$$

Because $\operatorname{div}(f\nabla u) = 1$, convolution commutes with differentiation in the interior and gives

$$\mathcal{R}_h = -\operatorname{div}\left((S_h f)(S_h \nabla u) - S_h(f \nabla u)\right).$$

Both f and ∇u have uniformly bounded C^α-norms. Lemma 2.2 therefore gives, with

$$B_h = h^\alpha L_h^2,$$

the bounds

$$\|\mathcal{R}_h\|_2 \le C B_h, \qquad \|\mathcal{R}_h\|_{H^1} \le C h^{-1} B_h. \tag{25}$$

Also, Lemmas 2.2 and 2.3 give

$$\|E_h a - f\|_{L^2(D)} \le C h^\alpha L_h. \tag{26}$$

Control of the nonlinear noise contribution

All identities in this paragraph are on $\mathcal{A}_h$. Put

$$e = \widetilde{f}_h - a, \qquad S_Z b = \operatorname{div}(b \nabla Z_h).$$

Since $T_{\widehat{u}_h} = T_v + S_Z$,

$$T_{\widehat{u}_h} e = \mathcal{R}_h - S_Z a. \tag{27}$$

The elementary product bounds

$$\|S_Z a\|_2 \le C \|Z_h\|_{H^2}, \qquad \|S_Z a\|_{H^1} \le C\big(\|Z_h\|_{H^3} + h^{-1}\|Z_h\|_{H^2}\big)$$

follow by differentiating $a\Delta Z_h + \nabla a \cdot \nabla Z_h$ and using the displayed bounds on a. Lemma 3.1, the definition of $\mathcal{A}_h$, and (25) yield the rough estimates

$$\|e\|_2 \le C\big(B_h + \|Z_h\|_{H^2}\big), \tag{28}$$

$$\|e\|_{H^1} \le C\big(h^{-1} B_h + \|Z_h\|_{H^3} + h^{-1}\|Z_h\|_{H^2}\big). \tag{29}$$

To obtain the sharp variance, rewrite (27) as

$$e = P_v \mathcal{R}_h - P_v S_Z a - P_v S_Z e. \tag{30}$$

The middle term is linear in the noise and, by Lemma 4.2,

$$\mathbb{E}\|P_v S_Z a\|_2^2 \le C n^{-1} h^{-d-3}. \tag{31}$$

The last term is controlled without any independence assumption between the estimated flow and the noise. Namely,

$$\|P_v S_Z e\|_2 \le C\left(\|Z_h\|_{1,\infty}\|e\|_{H^1} + \|Z_h\|_{2,\infty}\|e\|_2\right).$$

Substitute (28)–(29) and apply Lemma 4.1 and Cauchy–Schwarz. This gives

$$\mathbb{E}\left[\mathbf{1}_{\mathcal{A}_h}\|P_v S_Z e\|_2^2\right] \le C\left(B_h^2 n^{-1} h^{-d-4} L_h + n^{-2} h^{-2d-8} L_h\right). \tag{32}$$

For clarity, a representative fourth-moment calculation is

$$\mathbb{E}\left(\|Z_h\|_{1,\infty}^2 \|Z_h\|_{H^3}^2\right) \le \left(\mathbb{E}\|Z_h\|_{1,\infty}^4\right)^{1/2} \left(\mathbb{E}\|Z_h\|_{H^3}^4\right)^{1/2} \le C n^{-2} h^{-2d-8} L_h.$$

The term with $h^{-1}\|Z_h\|_{1,\infty}\|Z_h\|_{H^2}$ has the same bound, as does the term with $\|Z_h\|_{2,\infty}\|Z_h\|_{H^2}$. The terms containing B_h give the first term in (32).

Combining (30)–(32),

$$\mathbb{E}[\mathbf{1}_{\mathcal{A}_h}\|e\|_2^2] \le C\left(B_h^2 + n^{-1}h^{-d-3} + B_h^2 n^{-1} h^{-d-4} L_h + n^{-2} h^{-2d-8} L_h\right).$$

Our bandwidth satisfies $n^{-1} = h^{2\alpha+d+3}$. Hence the last display becomes

$$C\left(h^{2\alpha} L_h^4 + h^{2\alpha} + h^{2\alpha} L_h^4 h^{2\alpha-1} L_h + h^{4\alpha-2} L_h\right) \le C h^{2\alpha} L_h^4.$$

Here $\alpha > 1$ is used in $h^{4\alpha-2} L_h \le C h^{2\alpha}$.

Projection onto $[1/2, 3/2]$ cannot increase the loss, since the true conductivity takes values in that interval. By (10), (26), and (24),

$$\sup_{f \in \mathcal{F}(\alpha, r_*)} \mathbb{E}_f \|\widehat{f}_n - f\|_2^2 \le C h^{2\alpha} L_h^4 + C n^{-3}.$$

This is the asserted upper bound.

If $\alpha \notin \mathbb{N}$, use the versions of the approximation lemmas without logarithms, so that $B_h = h^\alpha$. The logarithms remaining in the stochastic remainder are absorbed by the strictly positive powers $h^{2\alpha-1}$ and $h^{2\alpha-2}$. The resulting upper bound is $Ch^{2\alpha}$. □

6 The lower bound

We construct pressure perturbations with fine angular oscillations and fixed radial scale.

6.1 Angular bumps

Let $m = d-1$. Fix a smooth coordinate chart $\chi : Q \to \mathbb{S}^{d-1}$, and a smaller cube compactly contained in Q. For sufficiently small h, choose centers $z_1, \dots, z_{M_h}$ in this cube, separated by $4h$, such that

$$c h^{-m} \le M_h \le C h^{-m}.$$

Fix a nonzero $\psi \in C_c^\infty((-1,1)^m)$, and define

$$\psi_{j,h}(\chi(z)) = \psi((z - z_j)/h),$$

extended by zero outside the chart. Their supports are pairwise disjoint. Coordinate changes on the fixed chart give

$$\|\psi_{j,h}\|^2_{L^2(\mathbb{S}^{d-1})} \asymp h^m, \tag{33}$$

$$\|\nabla_{\mathbb{S}^{d-1}}\psi_{j,h}\|^2_{L^2(\mathbb{S}^{d-1})} \asymp h^{m-2}. \tag{34}$$

Integration by parts and Cauchy–Schwarz imply

$$\|\Delta_{\mathbb{S}^{d-1}}\psi_{j,h}\|_2^2 \geq \frac{\|\nabla_{\mathbb{S}^{d-1}}\psi_{j,h}\|_2^4}{\|\psi_{j,h}\|_2^2} \geq ch^{m-4}. \tag{35}$$

For $\theta \in \{0,1\}^{M_h}$, write $q_\theta = \sum_j \theta_j \psi_{j,h}$. Disjointness gives

$$\|q_\theta\|_2^2 \leq C, \qquad \|\nabla^k_{\mathbb{S}^{d-1}} q_\theta\|_\infty \leq C_k h^{-k}$$

for every fixed k.

6.2 Realization as conductivities

Fix a nonnegative, nonzero $\rho \in C_c^\infty((1/4,1/2))$. For a fixed small $\kappa > 0$, define on $\mathbb{R}^d$

$$v_\theta(x) = \kappa h^{\alpha+2}\rho(|x|)q_\theta\left(\frac{x}{|x|}\right), \qquad u_\theta = u_\circ + v_\theta.$$

The perturbation is zero near 0, so this defines a smooth function everywhere. For each fixed k,

$$\|\nabla^k v_\theta\|_\infty \leq C_k \kappa h^{\alpha+2-k}. \tag{36}$$

In particular, for sufficiently small κ, $D^2 u_\theta$ is uniformly positive definite, independently of h, θ.

Lemma 6.1 (Conductivity realization and expansion)**.** *For sufficiently small κ, there are smooth positive conductivities $f_\theta = 1 + g_\theta$ such that*

$$\operatorname{div}(f_\theta \nabla u_\theta) = 1 \quad \text{on } \mathbb{R}^d,$$

and, uniformly in h, θ,

$$\|\nabla^k g_\theta\|_\infty \leq C_k \kappa h^{\alpha-k} \quad \text{for every fixed } k, \tag{37}$$

$$\|g_\theta\|_{C^\alpha(D)} \leq C\kappa, \tag{38}$$

$$g_\theta = \ell_\theta + \varepsilon_\theta, \qquad \ell_\theta = -P_{u_\circ}\Delta v_\theta, \qquad \|\varepsilon_\theta\|_{L^2(D)} \leq C\kappa^2 h^{2\alpha}. \tag{39}$$

Moreover, $u_{f_\theta} = u_\theta$ on D.

Proof. The backward-gradient flow of u_θ is globally defined on $\mathbb{R}^d$: its vector field is globally Lipschitz and has at most linear growth. Its unique critical point is 0, and the flow contracts towards 0. Define $f_\theta = P_{u_\theta}\mathbf{1}$ by the same integral as (14). That formula and its proof apply on $\mathbb{R}^d$ here, since the Hessian has uniform positive bounds and all derivatives of order at least two are bounded. Thus f_θ is smooth, positive, bounded, and satisfies the stated equation.

Write $u = u_\theta$, $v = v_\theta$, and $g = f_\theta - 1$. Then

$$T_u g = -\Delta v. \tag{40}$$

The L^∞ transport estimate gives $\|g\|_\infty \le C\kappa h^\alpha$.

We prove (37) by induction. Differentiating (40) k times, and collecting the terms containing k-th derivatives of g, gives

$$T_u(\nabla^k g) + H_k \nabla^k g = F_k.$$

Here H_k is the sum of the actions of D^2u on the k tensor indices. If $D^2u \ge \lambda I$, it is symmetric and bounded below by $k\lambda I$. Leibniz's rule gives

$$\begin{aligned}\|F_k\|_\infty \le C_k \Bigg(& \|\nabla^{k+2} v\|_\infty \\ & + \sum_{j=2}^{k} \|\nabla^{j+1} v\|_\infty \|\nabla^{k-j+1} g\|_\infty \\ & + \sum_{j=1}^{k} \|\nabla^{j+2} v\|_\infty \|\nabla^{k-j} g\|_\infty \Bigg).\end{aligned}$$

All derivatives of $u_\circ$ of order at least three vanish. Along each trajectory, the homogeneous matrix equation has propagator norm at most $e^{-(d+k)\lambda t}$. Consequently

$$\|\nabla^k g\|_\infty \le ((d+k)\lambda)^{-1} \|F_k\|_\infty.$$

The differentiability and boundedness needed here also follow by the variational-equation argument in Lemma 3.1. Using the induction hypothesis and (36), the right side is bounded by

$$C_k\big(\kappa h^{\alpha-k} + \kappa^2 h^{2\alpha-k}\big) \le C_k' \kappa h^{\alpha-k}.$$

This proves (37).

Choose an integer $J > \alpha$. The dyadic characterization implies

$$\|g\|_{C^\alpha(\mathbb{R}^d)} \le C \left(\|g\|_\infty + \|g\|_\infty^{1-\alpha/J} \|\nabla^J g\|_\infty^{\alpha/J} \right).$$

Indeed, a dyadic block is bounded both by $C\|g\|_\infty$ and, using its vanishing moments, by $C 2^{-jJ} \|\nabla^J g\|_\infty$; optimize between these bounds. Substitution of (37) proves (38), including at integer α.

Expanding the original equation around $(1, u_\circ)$ gives

$$T_{u_\circ} g = -\Delta v - \operatorname{div}(g \nabla v).$$

Therefore

$$g = -P_{u_\circ} \Delta v - P_{u_\circ} \operatorname{div}(g \nabla v).$$

The L^∞ bound for $P_{u_\circ}$, together with (36) and (37) at orders zero and one, yields

$$\|P_{u_\circ} \operatorname{div}(g\nabla v)\|_\infty \le C \left(\|\nabla g\|_\infty \|\nabla v\|_\infty + \|g\|_\infty \|\Delta v\|_\infty \right) \le C\kappa^2 h^{2\alpha}.$$

This proves (39).

Finally, v_θ vanishes near ∂D, so $u_\theta = 0$ there. Since $f_\theta > 0$, uniqueness of the Dirichlet problem implies $u_{f_\theta} = u_\theta$. □

6.3 Separation of the conductivities

For $x = r\omega$, the unperturbed inverse is explicitly

$$P_{u_\circ} s(r\omega) = d r^{-d} \int_0^r t^{d-1} s(t\omega)\, dt.$$

Using the polar-coordinate Laplacian, for $r > 1/2$ we obtain

$$\ell_\theta(r\omega) = -d\kappa h^{\alpha+2} J_\rho r^{-d} \Delta_{\mathbb{S}^{d-1}} q_\theta(\omega), \qquad J_\rho = \int_0^\infty t^{d-3} \rho(t)\, dt > 0. \tag{41}$$

Indeed, the radial part integrates to $[t^{d-1}\rho'(t)]_0^r = 0$.

Let $H(\theta, \eta)$ be Hamming distance. Since the angular supports are disjoint, integration over the fixed annulus $3/4 < |x| < 7/8$, using (35) and (41), gives

$$\|\ell_\theta - \ell_\eta\|^2_{L^2(D)} \geq c\kappa^2 h^{2\alpha+m} H(\theta, \eta). \tag{42}$$

There is a subset $\Theta_h \subset \{0,1\}^{M_h}$ such that

$$\log |\Theta_h| \geq c M_h, \qquad H(\theta, \eta) \geq M_h/8 \quad (\theta \neq \eta). \tag{43}$$

This is the usual binary packing bound; see, for example, [4, Lemma 2.9]. One elementary proof is to select words greedily, removing at each step a Hamming ball of radius $\lfloor M_h/8 \rfloor$. Such a ball has at most

$$\sum_{k \leq M_h/8} \binom{M_h}{k} \leq \exp\{M_h H_b(1/8)\}$$

points, where $H_b(t) = -t \log t - (1-t)\log(1-t) < \log 2$. The binomial bound follows by multiplying by $(t/(1-t))^k$ and applying the binomial theorem. The greedy procedure therefore leaves exponentially many selected words and proves (43).

Since $M_h \asymp h^{-m}$, equations (42) and (43) imply

$$\|\ell_\theta - \ell_\eta\|_2 \geq c\kappa h^\alpha \quad (\theta, \eta \in \Theta_h,\ \theta \neq \eta).$$

Lemma 6.1 then gives, for all sufficiently small h,

$$\|f_\theta - f_\eta\|_2 \geq c' \kappa h^\alpha. \tag{44}$$

The remainder is negligible because $\kappa^2 h^{2\alpha} = o(\kappa h^\alpha)$.

6.4 Testing bound

Proposition 6.2. *For every fixed sufficiently small $r > 0$,*

$$R_n(\alpha, r) \geq c\, n^{-2\alpha/(2\alpha+d+3)}$$

for all sufficiently large n.

Proof. Choose a fixed positive κ so small that all the preceding constructions apply and $C\kappa \leq r$, where C is the constant in (38). Thus every f_θ belongs to $\mathcal{F}(\alpha, r)$.

Let P_θ be the observation law under f_θ, and let $P_\circ$ be that under $f = 1$. For Gaussian white noise with variance level n^{-1},

$$\mathrm{KL}(P_\theta, P_\circ) = \frac{n}{2}\|u_\theta - u_\circ\|_2^2 = \frac{n}{2}\|v_\theta\|_2^2 \le Cn\kappa^2 h^{2\alpha+4}. \tag{45}$$

The equality follows directly from the Gaussian likelihood

$$\frac{dP_\theta}{dP_\circ} = \exp\left\{n\big(Y_n(v_\theta) - \langle u_\circ, v_\theta\rangle\big) - \frac{n}{2}\|v_\theta\|_2^2\right\}.$$

It can equivalently be obtained by finite-dimensional projections and convergence of the isonormal series. The inequality in (45) uses $\|q_\theta\|_2^2 \le C$.

Take

$$h = n^{-1/(2\alpha+d+3)}.$$

Since $m = d - 1$,

$$nh^{2\alpha+4} = h^{-m}, \qquad \log|\Theta_h| \ge ch^{-m}.$$

Decrease the fixed κ, if necessary, so that

$$\frac{1}{|\Theta_h|}\sum_{\theta\in\Theta_h} \mathrm{KL}(P_\theta, P_\circ) \le \frac{1}{8}\log|\Theta_h|. \tag{46}$$

We recall the finite-hypothesis information inequality used here, a form of Fano's inequality; compare [4, Theorem 2.5 and Chapter 2]. Let J be uniform over N hypotheses and let $\overline{P} = N^{-1}\sum_j P_j$. For every reference law P_0,

$$I(J;Y) = \frac{1}{N}\sum_j \mathrm{KL}(P_j, \overline{P}) \le \frac{1}{N}\sum_j \mathrm{KL}(P_j, P_0),$$

because the difference of the right side and the middle expression is $\mathrm{KL}(\overline{P}, P_0)$. For any classifier $\widehat{J}$, with error probability p_e, the binary error indicator gives

$$H(J \mid Y) \le \log 2 + p_e \log N.$$

Since $H(J) = \log N$, it follows that

$$p_e \ge 1 - \frac{I(J;Y) + \log 2}{\log N}.$$

Applying this with (46) shows that every classifier has average error probability at least $1/2$ for large n.

Given any $L^2(D)$-valued estimator, classify by its nearest member of $\{f_\theta : \theta \in \Theta_h\}$, resolving ties by a fixed rule. By (44), misclassification implies an L^2-error of at least $c\kappa h^\alpha/2$. Consequently,

$$\sup_{f\in\mathcal{F}(\alpha,r)} \mathbb{E}_f\|\widehat{f} - f\|_2^2 \ge \frac{1}{|\Theta_h|}\sum_{\theta\in\Theta_h} \mathbb{E}_\theta\|\widehat{f} - f_\theta\|_2^2 \ge c\kappa^2 h^{2\alpha}.$$

Taking the infimum over estimators proves the proposition. □

Proof of Theorem 1.1. Choose r_* sufficiently small for Lemma 2.1 and Proposition 5.1. The upper bound then holds for every subclass $\mathcal{F}(\alpha, r)$, $0 < r \le r_*$. Proposition 6.2, with its fixed κ chosen according to r, gives the matching lower bound. □

Remark 6.3. The effective denominator is $2\alpha + d + 3$, rather than the denominator obtained by treating the forward map as an isotropic operator of a single differential order. In the lower bound there are $d - 1$ angular directions and a two-derivative attenuation. In the upper bound, transport integration reduces the integrated variance of a twice differentiated, smoothed white noise from order $n^{-1}h^{-d-4}$ to order $n^{-1}h^{-d-3}$. The small-radius assumption supplies uniform convexity of the pressure and uniform control of this transport geometry.

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